Volume Builder

TEACHER KEY

5th Grade Math · Interactive Math · Explore & Reflect

Get started

Stack unit cubes into rectangular prisms, see layers and base area, and connect counting cubes to V = l × w × h and composite figures. Open the tool and use it to answer each question.

Explore

1. Build 4 × 3 × 2. How many cubes are in the bottom layer?
   12
2. What is the volume of 4 × 3 × 2?
   24 cubic units
3. Find two different prisms with volume 36.
   Examples: 6 × 3 × 2, 4 × 3 × 3
4. A prism has base area 15 and height 4. What is its volume?
   60 cubic units
5. Build a composite figure: Prism A 3 × 2 × 4 and Prism B 2 × 2 × 2. Total volume?
   32 cubic units

Volume Builder

TEACHER KEY

5th Grade Math · Interactive Math · Explore & Reflect

Explore more

6. Why do we use cubic units for volume?

Volume counts cubes that fill space in three directions.

7. If you double the height, what happens to the volume?

It doubles.

Reflect

8. Reflect: Explain why l × w × h counts every cube, even hidden ones.

l × w counts one full layer, and h counts how many layers are stacked.

Teacher notes

Accept equivalent forms (for example 1/2 = 2/4 = 0.5) and any correct example where several answers are possible. Have students explain using the tool's picture.